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RESEARCH ARTICLE

Linaclotide in Chronic Idiopathic Constipation Patients with Moderate to Severe Abdominal Bloating: A Randomized, Controlled Trial

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Abstract

Background

Abdominal bloating is a common and bothersome symptom of chronic idiopathic constipation. The objective of this trial was to evaluate the efficacy and safety of linaclotide in patients with chronic idiopathic constipation and concomitant moderate-to-severe abdominal bloating.

Methods

This Phase 3b, randomized, double-blind, placebo-controlled clinical trial randomized patients to oral linaclotide (145 or 290 μg) or placebo once daily for 12 weeks. Eligible patients met Rome II criteria for chronic constipation upon entry with an average abdominal bloating score $\geq$5 (self-assessment: 0 10-point numerical rating scale) during the 14-day baseline period. Patients reported abdominal symptoms (including bloating) and bowel symptoms daily; adverse events were monitored. The primary responder endpoint required patients to have $\geq$3 complete spontaneous bowel movements/week with an increase of $\geq$1 from baseline, for $\geq$9 of 12 weeks. The primary endpoint compared linaclotide 145 μg vs. placebo.

Results

The intent-to-treat population included 483 patients (mean age=47.3 years, female=91.5%, white=67.7%). The primary endpoint was met by 15.7% of linaclotide 145 μg patients vs. 7.6% of placebo patients (P<0.05). Both linaclotide doses significantly improved abdominal bloating vs. placebo (P<0.05 for all secondary endpoints, controlling for multiplicity). Approximately one-third of linaclotide patients (each group) had $\geq$50% mean decrease

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from baseline in abdominal bloating vs. 18% of placebo patients (P<0.01). Diarrhea was reported in 6% and 17% of linaclotide 145 and 290 μg patients, respectively, and 2% of placebo patients. AEs resulted in premature discontinuation of 5% and 9% of linaclotide 145 μg and 290 μg patients, respectively, and 6% of placebo patients.

Conclusions

Once-daily linaclotide (145 and 290 μg) significantly improved bowel and abdominal symptoms in chronic idiopathic constipation patients with moderate-to-severe baseline abdominal bloating; in particular, linaclotide significantly improved abdominal bloating compared to placebo, an important finding given the lack of agents available to treat abdominal bloating in chronic idiopathic constipation patients.

Trial Registration

ClinicalTrials.gov NCT01642914

Introduction

Chronic idiopathic constipation (CIC), estimated to affect between 12% and 19% of North Americans,[1] is characterized by a variety of bowel symptoms including reduced bowel movement (BM) frequency, hard stools, straining during defecation, and a sense of incomplete evacuation after defecation, as well as abdominal symptoms of bloating and discomfort.[2–4] Abdominal bloating refers to subjective sensations of excessive gas, a fullness or tightness in the abdomen, or a feeling of abdominal swelling.[5,6] Subjective sensations of abdominal bloating may or may not be associated with distention, which can be defined as a visible change in abdominal girth.[6] Abdominal bloating is a common and particularly bothersome symptom of CIC.[4,6,7] The etiology of abdominal bloating is incompletely understood, may differ from patient to patient, and is likely multifactorial in nature.[8] Potential causes include visceral hypersensitivity, abnormal intestinal gas transit, impaired evacuation of rectal gas, excess fermentation of bowel contents, an abnormal abdomino-diaphragmatic reflex, and disorders related to the gut microbiota.[6,9] Few treatments have been shown to improve abdominal bloating in patients with CIC.[10,11] In fact, abdominal bloating may be exacerbated by some treatments aimed at improving constipation-related symptoms.[10,11]

Linaclotide is a minimally absorbed 14-amino-acid peptide which binds to and activates guanylate cyclase C (GC-C) on the luminal surface of the intestinal epithelium. Activation of intestinal GC-C receptors results in increased cyclic guanosine monophosphate (cGMP) production, which in turn causes chloride and bicarbonate to be secreted into the gastrointestinal lumen, with consequent increased fluid secretion and accelerated intestinal transit.[12–14] Linaclotide reduces visceral hypersensitivity in animal models, an effect that may be related to cGMP modulation of afferent nerve activity.[15–17] The results of two published Phase 3 trials of linaclotide in CIC patients demonstrated the effectiveness of linaclotide at improving the bowel and abdominal symptoms of CIC in both men and women.[18] Linaclotide is approved in the United States (US), Canada, and Mexico at a dose of 145 μg once daily for the treatment of CIC in adult men and women.

The objective of this Phase 3b clinical trial was to assess the efficacy and safety of linaclotide, at doses of 145 μg and 290 μg administered once daily, in patients with CIC and moderate to severe abdominal bloating at baseline. The hypothesis that linaclotide treatment would lead to an improvement in patient self-assessments of abdominal bloating was of specific interest and will be the key focus of this paper.

Methods

Design Overview

This 12-week, multicenter, randomized, double-blind, placebo-controlled, parallel-group, multiple fixed-dose trial was conducted at 141 clinical sites (136 in the United States and 5 in Canada). Recruitment began in August 2012 and ended in February 2013, when the planned sample size was reached. The trial was conducted between 17 August 2012 (first patient enrolled) and 29 May 2013 (last patient completed). The trial was designed, conducted, and reported in compliance with the ethical principles that have their origins in the Declaration of Helsinki and the principles of Good Clinical Practice guidelines. An Institutional Review Board approved the protocol and all trial procedures for all trial centers (Quorum Review IRB, Western Institutional Review Board [WIRB], Mercy Medical Center Institutional Review Board, University of Oklahoma Health Sciences Center Institutional Review Board). All patients gave written informed consent prior to participating in any trial-related procedures. The trial is registered at ClinicalTrials.gov (registration number NCT01642914).

The randomization sequence was generated in SAS by a statistical programmer not otherwise assigned to this trial and the sequence was provided securely to the interactive voice response system (IVRS) vendor. Randomization was blocked by dynamically allocating blocks of 6 to clinical sites as needed; each block contained a random sequence of treatments (145 μg or 290 μg of linaclotide or matching placebo) at a 1:1:1 ratio. Randomization numbers within each block were assigned in ascending order and additional blocks were allocated to clinical sites once complete blocks were used. Trial personnel used the IVRS to randomize patients and to obtain instructions on the study drug to be dispensed.

During an initial screening period of up to 21 days, patients provided blood and urine samples for laboratory testing and were instructed to discontinue any prohibited medications (including laxatives, suppositories, or enemas used to treat CIC). Patients meeting the inclusion and exclusion criteria then entered a 14- to 21-day pretreatment baseline period during which they provided daily and weekly symptom assessments via IVRS. Subsequently, eligible patients were randomized to receive 145 μg or 290 μg of linaclotide or matching placebo (identical appearance and containing the same inactive substances), as an oral capsule administered once daily, for the duration of the 12-week treatment period. Patients were instructed to take the study drug at least 30 minutes before breakfast each day and to make daily IVRS calls to report their symptoms throughout the treatment period.

Trial visits were scheduled at screening, at the start of the baseline period, at randomization (day 1), and throughout the treatment period (weeks 2, 4, 8, and 12 [end-of-trial visit]). All patients and personnel involved in the design and implementation of the trial remained blinded to treatment assignments until the database was locked.

Trial Patients

Male and female patients were eligible to participate if they were at least 18 years of age and met modified Rome II criteria for chronic constipation.[19] The criteria included < 3 spontaneous bowel movements (SBMs: BMs occurring in the absence of laxative, suppository, or enema use during the preceding 24 hours) per week and at least one of the following symptoms during $> 25\%$ of BMs for at least 12 weeks (not necessarily consecutive) within the preceding 12 months: straining, lumpy or hard stools, and a sensation of incomplete evacuation.